HEATSAFE PAKISTAN

City-scale heat intelligence for safer outdoor-work planning

HackInsight 2026 | Climate and heat-health | Summary (1 page)

16	1.056 C	4.65%	21
CITIES IN FEED	TEST MODEL MAE	TEST MAE GAIN	EXPORTED RUN RECORDS

PROBLEM AND USER

Outdoor workers and construction-site supervisors in Pakistan need timely, understandable heat information for planning breaks, awareness, and workplace risk assessments. HeatSafe provides forecasts with uncertainty, cited safety guidance, and data freshness information, rather than claiming to diagnose or certify workplace safety.

DATA AND TRUSTWORTHY ENGINEERING

Historical Karachi weather: 52,608 ERA5 hourly records (2020-2025). CAMS modelled air-quality data: 26,304 records (2023-2025). Raw, cleaned and processed/gold DuckDB layers, source manifests and Data Quality Agent fix/fill/flag decision logs are part of the architecture. A separate 48-hour Open-Meteo provider forecast feed now covers 16 Pakistani cities (768 records); it is not a city-trained ML forecast.

MODEL: DECLARED BASELINE AND HISTORICAL HELD-OUT EVALUATION

A HistGradientBoosting model predicts weather-derived heat index 24 hours ahead for Karachi. Time-only split: train 2020-2023, validate Jan-Jun 2024, calibrate Jul-Dec 2024, test 2025 (8,760 rows). Test MAE: 1.05595 C versus previous-day persistence baseline 1.10741 C (4.65% improvement); R2: 0.9683. Nominal 90% historical interval: +/-3.00655 C; empirical test coverage: 95.38%. Derived risk-category accuracy: 87.41%.

AGENTS, AUTOMATION AND USER EXPERIENCE

The Data Quality Agent records rule-level fix, fill or flag decisions. The RAG agent retrieves official CDC NIOSH, UK HSE and NOAA guidance, and Gemini generates cited general explanations with refusals for unsupported facts; live Karachi numerical forecasts come from DuckDB gold, not Gemini. Prefect scheduled hourly runs completed, with five green tasks shown at 2 PM, 3 PM, 4 PM and 5 PM on Oct 21; 21 run records were exported. A teammate-built web dashboard is being integrated.

VALUE AND RESPONSIBLE LIMITS

Primary users: site supervisors, safety officers and outdoor workers. Potential sustainability: employer subscriptions or funded safety pilots, subject to validation; running costs include hosting, APIs, maintenance and data review. Heat index is not measured WBGT or worker illness, and the Karachi model is not yet validated for other cities. Cite gold rows and official documents; do not treat this as clinical or worksite-specific safety advice.

Evidence: modeling/evaluations/*.json | logs/ | evidence/prefect_runs.csv | manifest.csv

Source repo: github.com/Hamdanilyas01/Hack_insight- | Review live deployment and final scope before submitting.